

Day 02 – Networking

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1-theory

1. What is a computer network?

A computer network is a group of connected devices that communicate and share data and resources.

2. What is the difference between a client and a server?

A client requests a service; a server provides the service or resource.

3. What is a LAN?

A Local Area Network connects devices in a small area such as a home, school, or office.

4. What is a WAN?

A Wide Area Network connects networks over large geographical areas.

The Internet is an example.

5. What is the purpose of a switch?

A switch connects devices on a LAN and forwards data to the correct device.

6. What is the purpose of a router?

A router connects different networks(LANS) and forwards packets between them.

7. What is a firewall?

A firewall monitors and controls network traffic based on security rules.

8. What is an IP address?

An IP address is a logical address used to identify a device/interface on a network.

9. What is a MAC address?

A MAC address is a hardware-level address used to identify a network interface on a local network.

10. What is a default gateway?

The default gateway is the device, usually a router, that sends traffic to other networks.

11. What is DNS?

DNS (Domain Name System) translates domain names such as google.com into IP addresses.

12. Why is networking knowledge important for cybersecurity?

It helps security professionals understand traffic, detect attacks, investigate incidents, and protect networks

2. My network information

IPv4 Address: 192.168.100.209

Subnet Mask: 255.255.255.0
Default Gateway: 192.168.100.1
DNS Server: 192.168.100.1
MAC / Physical Address: C4-13-75-3C-F1-01

3.screenshots

1.ipconfig

```
Windows PowerShell
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PS C:\Windows\System32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::e3bb:c8ce:a438:c2e7%6
    IPv4 Address. . . . . : 192.168.100.209
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

2.tracert

```
PS C:\Users\perehan_sewid> tracert google.com

Tracing route to google.com [172.217.23.174]
over a maximum of 30 hops:

  1     3 ms     1 ms     1 ms  dev.opt [192.168.100.1]
  2     6 ms     4 ms     3 ms  100.127.132.1
  3     4 ms     7 ms     3 ms  10.45.10.53
  4     4 ms     6 ms     5 ms  10.35.14.2
  5     5 ms    37 ms     6 ms  10.45.28.66
  6    38 ms    37 ms    38 ms  72.14.205.114
  7    45 ms    45 ms    44 ms  fra15s22-in-f174.1e100.net [172.217.23.174]

Trace complete.
PS C:\Users\perehan_sewid> |
```

3.ping

```
PS C:\Windows\System32> ping google.com

Pinging google.com [172.217.23.174] with 32 bytes of data:
Reply from 172.217.23.174: bytes=32 time=48ms TTL=114
Reply from 172.217.23.174: bytes=32 time=44ms TTL=114
Reply from 172.217.23.174: bytes=32 time=44ms TTL=114
Reply from 172.217.23.174: bytes=32 time=45ms TTL=114

Ping statistics for 172.217.23.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 48ms, Average = 45ms
PS C:\Windows\System32> |
```